The compact-support fractional Poincaré inequality at every $1 \leq p < \infty$

Full-resolution packet for `fa_banach_001`

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Status. Candidate full resolution of Question 7.4 in arXiv:2001.06210, subject to human review. The range $1 < p < \infty$ appeared later in the literature. The proof below also covers the omitted endpoint $p = 1$, answers the authors' lower-order-derivative question for every $0 \leq t \leq s$, and gives the requested dependence on a classical Poincaré constant when $1 < p < \infty$ and $s \geq 1$.

Source question

G. Covi, K. Mönkkönen, and J. Railo, *Unique continuation property and Poincaré inequality for higher order fractional Laplacians with applications in inverse problems*, arXiv:2001.06210 [1]. Question 7.4 on PDF page 32 asks whether, for $s \geq 0$, $1 \leq p < \infty$, compact $K \subset \mathbb{R}^n$, and $u \in H^{s,p}(\mathbb{R}^n)$ supported in K,

$$\|u\|_{L^p(\mathbb{R}^n)} \leq C(n, K, s, p)\|(-\Delta)^{s/2}u\|_{L^p(\mathbb{R}^n)}. \tag{1}$$

The following paragraph asks whether u can be replaced by $(-\Delta)^{t/2}u$, $0 \leq t \leq s$, and whether the constant can be expressed using a classical Poincaré constant for $s \geq 1$.

problems. The UCPs can be used to show Runge approximation properties for nonlocal equations such as the fractional Schrödinger equation (see theorem 1.7), which in turn can be used to show uniqueness for the corresponding nonlocal inverse problem (see theorem 1.6). The UCPs have also applications in integral geometry, where the uniqueness of the ROI problem for the d-plane transform can be reduced to a unique continuation problem for the fractional Laplacian (see remark 4.1 and corollaries 1.3 and 1.4).

7.2. **Fractional Poincaré inequality for L^p-norms.** In section 3.2 we prove the fractional Poincaré inequality for L^2-norms in multiple ways. The inequality is needed for the well-posedness of the inverse problem for the fractional Schrödinger equation. One could try to extend the Poincaré inequality for general L^p-norms. This suggests the following natural question which is also interesting from the pure mathematical point of view.

Question 7.4. *Let $s \geq 0$, $1 \leq p < \infty$, $K \subset \mathbb{R}^n$ compact set and $u \in H^{s,p}(\mathbb{R}^n)$ such that $\mathrm{spt}(u) \subset K$. Show that there exists a constant $c = c(n, K, s, p)$ such that*

$$\|u\|_{L^p(\mathbb{R}^n)} \leq c \left\| (-\Delta)^{s/2} u \right\|_{L^p(\mathbb{R}^n)} \tag{27}$$

or give a counterexample.

Since we have presented several proofs for the Poincaré inequality in the case $p = 2$, one could try some of our methods to solve question 7.4. However, some of our proofs are heavily based on Fourier analysis and those approaches might be difficult to generalize to the L^p-case when $p \neq 2$. Like in theorem 1.5 and in theorem 3.17, another interesting question is whether one can replace u in the left-hand side of equation (27) with $(-\Delta)^{t/2}u$ when $0 \leq t \leq s$, and whether the constant c in equation (27) can be expressed in terms of the classical Poincaré constant when $s \geq 1$.

7.3. **The Calderón problem for determining a higher order PDO.** In this discussion, we try to make as simple assumptions as possible. The whole point is to introduce a new inverse problem that we think is a very natural and interesting one, at least from a pure mathematical point of view. Therefore the optimal regularity in the statement of the problem

Definitions and proof intuition

Write

$$\mathsf{D}^r = (-\Delta)^{r/2}, \qquad \mathsf{J}^r = (I - \Delta)^{r/2}, \qquad \widehat{\mathsf{D}^r u}(\xi) = |\xi|^r \widehat{u}(\xi).$$

The Bessel-potential space $H^{s,p}(\mathbb{R}^n)$ consists of $u = \mathsf{J}^{-s} f$ with $f \in L^p$, with norm $\|u\|_{H^{s,p}} = \|\mathsf{J}^s u\|_p$.

The usual Mikhlin and Riesz-transform arguments are not L^1-bounded. We instead use symbols whose inverse Fourier transforms are finite measures. They give the endpoint graph estimate

$$\|\mathsf{J}^s u\|_p \lesssim \|u\|_p + \|\mathsf{D}^s u\|_p \quad (1 \leq p \leq \infty).$$

If the Poincaré estimate failed, normalized counterexamples would be bounded in $H^{s,p}$. Fixed compact support and the L^1 Bessel kernel make this family precompact in L^p, even at $p = 1$. A limit would have Fourier transform supported at the origin, hence be a polynomial; no nonzero polynomial lies in finite L^p.

Endpoint-safe multiplier lemma

Let $\mathcal{W}(\mathbb{R}^n)$ be the Wiener algebra of Fourier transforms of finite complex Borel measures. If $m = \widehat{\mu} \in \mathcal{W}$, then $f \mapsto \mathcal{F}^{-1}(m\widehat{f}) = \mu * f$ is bounded on every L^p, $1 \leq p \leq \infty$.

Lemma 1 (Dyadic Wiener criterion)**.** *Let $N > n$, and let $a \in C^N(\mathbb{R}^n \setminus \{0\})$. Suppose that for all $|\alpha| \le N$,*

$$|\partial^\alpha a(\xi)| \le C_\alpha |\xi|^{\rho-|\alpha|} \quad (0 < |\xi| \le 1), \qquad |\partial^\alpha a(\xi)| \le C_\alpha |\xi|^{-\sigma-|\alpha|} \quad (|\xi| \ge 1)$$

for some $\rho, \sigma > 0$. Then $\mathcal{F}^{-1} a \in L^1(\mathbb{R}^n)$, hence $a \in \mathcal{W}(\mathbb{R}^n)$.

Proof. Choose a smooth annular dyadic partition $1 = \sum_{j\in\mathbb{Z}} \psi(2^{-j}\xi)$ away from zero and set $a_j(\xi) = a(\xi)\psi(2^{-j}\xi)$. Rescaling to a fixed annulus and integrating by parts $N > n$ times gives

$$\|\mathcal{F}^{-1} a_j\|_1 \le C \max_{|\alpha|\le N} 2^{j|\alpha|} \|\partial^\alpha a\|_{L^\infty(|\xi|\asymp 2^j)}.$$

The bound is $O(2^{j\rho})$ for $j \le 0$ and $O(2^{-j\sigma})$ for $j \ge 0$. Both series converge, so $\sum_j \mathcal{F}^{-1} a_j$ converges absolutely in L^1. □

Lemma 2 (The required symbols)**.** *For $s > 0$, the following symbols belong to $\mathcal{W}(\mathbb{R}^n)$:*

$$m_s(\xi) = \frac{(1+|\xi|^2)^{s/2}}{1+|\xi|^s}, \qquad b_s(\xi) = \frac{|\xi|^s}{(1+|\xi|^2)^{s/2}},$$
$$a_{s,t}(\xi) = \frac{|\xi|^t}{1+|\xi|^s}, \qquad 0 \le t \le s.$$

Proof. The symbol $m_s - 1$ satisfies Lemma 1 at both ends with positive exponent $\min\{s, 2\}$ (for $s = 2$ it vanishes). Hence $m_s \in \mathcal{W}$.

Let χ be smooth, compactly supported, and one near zero. The term χb_s has low-frequency exponent s, while $(1-\chi)(b_s - 1)$ has high-frequency decay exponent $\min\{s, 2\}$. These terms satisfy Lemma 1, and $b_s = \chi b_s + (1-\chi) + (1-\chi)(b_s - 1) \in \mathcal{W}$.

If $0 < t < s$, then $a_{s,t}$ has low exponent t and high decay exponent $s - t$, so the criterion applies. For $t = 0$, write $a_{s,0} = \chi + (a_{s,0} - \chi)$; the second term has exponent s at both ends. Finally $a_{s,s} = 1 - a_{s,0}$. □

Corollary 3 (Endpoint graph norm)**.** *For $s > 0$ and $1 \le p \le \infty$,*

$$\|\mathsf{J}^s u\|_p \le C_{n,s}\big(\|u\|_p + \|\mathsf{D}^s u\|_p\big). \tag{2}$$

Moreover $\mathsf{D}^s : H^{s,p} \to L^p$ is bounded, including at $p = 1$.

Proof. Use Lemma 2 in the identities

$$\widehat{\mathsf{J}^s u} = m_s(\widehat{u} + \widehat{\mathsf{D}^s u}), \qquad \widehat{\mathsf{D}^s u} = b_s \widehat{\mathsf{J}^s u}.$$

□

Full resolution

Theorem 4 (All-p compact-support Poincaré inequality)**.** *Let $K \subset \mathbb{R}^n$ be compact, $s > 0$, and $1 \le p < \infty$. There is $C = C(n, K, s, p)$ such that every $u \in H^{s,p}(\mathbb{R}^n)$ with* $\operatorname{supp} u \subset K$ *satisfies*

$$\|u\|_p \le C\|\mathsf{D}^s u\|_p. \tag{3}$$

More strongly, for every $0 \le t \le s$, there is $C_t = C(n, K, s, t, p)$ such that

$$\|\mathsf{D}^t u\|_p \le C_t \|\mathsf{D}^s u\|_p. \tag{4}$$

For $s = 0$, the source question is the identity with $C = 1$.

Proof. First we prove compactness at the endpoint. The Bessel kernel $G_s = \mathcal{F}^{-1}(1+|\xi|^2)^{-s/2}$ belongs to $L^1(\mathbb{R}^n)$. Indeed,

$$(1+|\xi|^2)^{-s/2} = \frac{1}{\Gamma(s/2)} \int_0^\infty e^{-r} r^{s/2-1} e^{-r|\xi|^2}\, dr$$

expresses it as a positive average of L^1-normalized Gaussian kernels. If $u = G_s * f$, then

$$\|u(\,\cdot+h) - u\|_p \le \|G_s(\,\cdot+h) - G_s\|_1 \|f\|_p = o_{h\to 0}(1)\|f\|_p.$$

Thus translations of the $H^{s,p}$ unit ball are uniformly small in L^p. Fixed compact support gives uniform tightness. The Fréchet–Kolmogorov criterion, valid for all $1 \le p < \infty$, proves

$$\{u \in H^{s,p} : \operatorname{supp} u \subset K\} \hookrightarrow L^p \quad \text{compactly.} \tag{5}$$

Suppose (3) fails. There are $u_j \in H^{s,p}$, supported in K, with

$$\|u_j\|_p = 1, \qquad \|\mathsf{D}^s u_j\|_p \longrightarrow 0.$$

Corollary 3 makes (u_j) bounded in $H^{s,p}$. By (5), after a subsequence $u_j \to u$ strongly in L^p, where $\operatorname{supp} u \subset K$ and $\|u\|_p = 1$.

Both convergences hold in tempered distributions. If $\varphi \in C_c^\infty(\mathbb{R}^n \setminus \{0\})$, then $\varphi/|\xi|^s$ is a smooth test function, and

$$\langle \widehat{u}, \varphi\rangle = \lim_j \langle \widehat{u_j}, \varphi\rangle = \lim_j \left\langle \widehat{\mathsf{D}^s u_j}, \frac{\varphi}{|\xi|^s} \right\rangle = 0.$$

Thus $\operatorname{supp} \widehat{u} \subset \{0\}$. A distribution supported at one point is a finite sum of derivatives of the Dirac mass, so its inverse Fourier transform is a polynomial. The only polynomial in $L^p(\mathbb{R}^n)$, $p < \infty$, is zero, contradicting $\|u\|_p = 1$. This proves (3).

Finally Lemma 2 and

$$\widehat{\mathsf{D}^t u} = a_{s,t}(\widehat{u} + \widehat{\mathsf{D}^s u})$$

give $\|\mathsf{D}^t u\|_p \le C(\|u\|_p + \|\mathsf{D}^s u\|_p)$. Apply (3) to obtain (4). □

Dependence on the classical Poincaré constant

Corollary 5. *Let $1 < p < \infty$, $s \ge 1$, and let Ω be a bounded Lipschitz domain with $K \Subset \Omega$. If*

$$\|v\|_{L^p(\Omega)} \le P_p(\Omega) \|\nabla v\|_{L^p(\Omega)} \quad (v \in W_0^{1,p}(\Omega)),$$

then, for $0 \le t \le s$,

$$\|\mathsf{D}^t u\|_p \le C(n,p,s,t) P_p(\Omega)^{s-t} \|\mathsf{D}^s u\|_p \tag{6}$$

for all $u \in H^{s,p}(\mathbb{R}^n)$ supported in K.

Proof. Riesz-transform boundedness gives $\|\nabla u\|_p \le C_{n,p}\|\mathsf{D}u\|_p$. For $s > 1$, the classical Poincaré and homogeneous interpolation inequalities give

$$\|u\|_p \le CP_p(\Omega)\|\mathsf{D}u\|_p \le CP_p(\Omega)\|u\|_p^{1-1/s}\|\mathsf{D}^s u\|_p^{1/s}.$$

Hence $\|u\|_p \le CP_p(\Omega)^s \|\mathsf{D}^s u\|_p$; for $s = 1$ this is immediate. Insert this into

$$\|\mathsf{D}^t u\|_p \le C\|u\|_p^{1-t/s}\|\mathsf{D}^s u\|_p^{t/s}$$

to obtain (6). The restriction $p > 1$ here reflects the failure of strong L^1 Riesz-transform boundedness; it is not needed in Theorem 4. □

Literature boundary and novelty check

Kar, Railo, and Zimmermann later stated this inequality on bounded sets as Theorem 3.1 of arXiv:2208.09528, published in *Calculus of Variations and Partial Differential Equations* 62 (2023) [2]. Their theorem, and the adjacent compact-embedding theorem, assume $1 < p < \infty$, not $p = 1$.

> 3.2. **Poincaré inequalities on Bessel potential spaces and the Rellich–Kondrachov theorem.** In this subsection we state a fractional Poincaré inequality and a variant of the Rellich–Kondrachov theorem which are adapted to our functional setting. The first result directly follows from Lemma 5.4 in [RZ22b]. The second one can be proved, as is done below, using compact embeddings in Besov-type and Triebel–Lizorkin-type spaces on smooth bounded domains [GHS21].
>
> **Theorem 3.1** (Fractional Poincaré inequality on bounded sets). *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set, $s > 0$, $1 < p < \infty$ and $\mathbb{K} = \mathbb{C}$ or $\mathbb{K} = \mathbb{R}^m$. Then there exists $C(n, p, s, \Omega, \mathbb{K}) > 0$ such that*
>
> $$\|u\|_{L^p(\mathbb{R}^n;\mathbb{K})} \leq C\|(-\Delta)^{s/2}u\|_{L^p(\mathbb{R}^n;\mathbb{K})}$$
>
> *for all $u \in \widetilde{H}^{s,p}(\Omega; \mathbb{K})$.*
>
> **Theorem 3.2** (Rellich–Kondrachov theorem). *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set, $s > 0$, $1 < p < \infty$ and $\mathbb{K} = \mathbb{C}$ or $\mathbb{K} = \mathbb{R}^m$. Then the embedding $\widetilde{H}^{s,p}(\Omega; \mathbb{K}) \hookrightarrow L^p(\mathbb{R}^n; \mathbb{K})$ is compact.*
>
> *Proof.* Without loss of generality we can restrict ourselves to the complex valued case. Let $\overline{\Omega} \subset \Omega'$ where Ω' is a smooth bounded domain. By [GHS21, Remark 2.6, Definition 2.10] one has $F^{s,0}_{p,q}(\Omega') = F^s_{p,q}(\Omega)$ for $s \in \mathbb{R}, 1 < p < \infty$, $0 < q \leq \infty$, where $F^{s,\tau}_{p,q}(\Omega)$, $0 \leq \tau \leq \infty$, denotes the generalized Triebel-Lizorkin space. Since the Triebel–Lizorkin spaces coincide with the Bessel potential space for $q = 2$ we have the identification $F^{s,0}_{p,2}(\Omega') = H^{s,p}(\Omega')$ for $s \in \mathbb{R}, 1 < p < \infty$. Therefore, [GHS21, Corollary 3.5] shows that $H^{s,p}(\Omega') \hookrightarrow L^p(\Omega')$ is compact. By the embeddings $\widetilde{H}^{s,p}(\Omega) \hookrightarrow \widetilde{H}^{s,p}(\Omega') \hookrightarrow H^{s,p}(\Omega')$ and $u = 0$ a.e. in $\mathbb{R}^n \setminus \overline{\Omega}$ for all $u \in \widetilde{H}^{s,p}(\Omega)$ with $s \geq 0$, it follows that $\widetilde{H}^{s,p}(\Omega) \hookrightarrow L^p(\mathbb{R}^n)$ is compact. $\square$
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> 3.3. **Caffarelli–Silvestre extension problems.** The purpose of this section is to recall the

A bounded search on 11–12 August 2026 checked the exact source title and arXiv id, Question 7.4, the later theorem title, and combinations of Bessel potential, fractional Poincaré, compact support, and endpoint $p = 1$. It recovered the source and the 2023 theorem but no explicit endpoint result or full all-t answer. The local run indexes contained no duplicate. Thus the $1 < p < \infty$ part is literature-known; the endpoint and unified proof are candidate-new, pending specialist review.

Verification notes and limitations

- No reflexivity, weak compactness, Mikhlin theorem, or strong L^1 Riesz-transform bound is used. Endpoint-sensitive steps use finite-measure multipliers and Fréchet–Kolmogorov compactness.
- The Fourier-support argument tests away from the origin, so it does not assume that $|\xi|^s$ is smooth at $\xi = 0$.
- The K-dependent constant in Theorem 4 is existential. Corollary 5 is quantitative for $p > 1$; no $P_1(\Omega)$-formula is asserted.
- The proof works for real or complex scalars and componentwise for finite-dimensional targets.

Human-review recommendation

Validity confidence is high. Review the dyadic L^1-kernel bound, the low/high asymptotics of the three radial symbols, and the endpoint Fréchet–Kolmogorov step. Novelty confidence is moderate because an endpoint result may be hidden under harmonic-analysis terminology; author and specialist citation review is recommended before publication claims.

References

[1] G. Covi, K. Mönkkönen, and J. Railo, *Unique continuation property and Poincaré inequality for higher order fractional Laplacians with applications in inverse problems*, arXiv:2001.06210 (2020).

[2] M. Kar, J. Railo, and P. Zimmermann, *The fractional p-biharmonic systems: optimal Poincaré constants, unique continuation and inverse problems*, arXiv:2208.09528; Calc. Var. Partial Differential Equations 62 (2023).